

Response to review comments — Vyuha Technical Document

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Point-by-point reply to the annotated review, mapped to the pages marked. Every figure is taken from the evaluation records in the repository, reported with confidence intervals; new results added since the review are collected in §7.4 of the technical document.

Thank you for the careful read and the encouragement. Below are point-by-point answers to each comment, mapped to the page you annotated. Every number cited is from the evaluation records in the repository (model card / paper), not restated from memory. I have also made two concrete edits to the document itself (noted at the end).

1. The QLoRA guard — "explain it more" and "is it not performing well?" (p7 underline, p13 chart)

The tuned QLoRA guard (Qwen2.5-1.5B + LoRA, 4-bit) is a **jailbreak detector**, and on that task it performs well: cross-benchmark **ROC-AUC 0.72-0.92 on jailbreak sets it never trained on** (out-of-distribution generalisation), at a false-refusal rate of 0.03-0.06. That is the "great detection rate" the text refers to.

The page-13 chart plotted only two axes — harmful-topic (XSTest) and semantic (PAIR) — where the guard scores 0.00 and 0.03. That is **inert by design**, not failure: detecting harmful topics and fluent persuasion is a different task, owned by the L2 content guard (Qwen3Guard), not by a jailbreak classifier. The chart made the guard look like a failure because it omitted the guard's own strong axis. **I have fixed the chart** — it now shows a second panel with the guard's OOD jailbreak ROC-AUC (0.72-0.92), clearly labelled as a different metric.

Honest caveat I want to state plainly: the QLoRA guard is the **most marginal component** in the stack. Its only justification is better OOD generalisation to novel jailbreak phrasings than the linear L1 detector gives. The paper already lists it under "components that did not clearly earn their place."

2. "Is it an existing technique?" (p13 circle)

Yes. QLoRA (Dettmers et al., 2023) is a standard fine-tuning method, and fine-tuning a guard classifier this way is common practice (Llama Guard, ShieldGemma are fine-tuned guards). What is *ours* is the specific trained adapter, the published artifact, and the measured finding that it behaves as a jailbreak specialist rather than a content guard — **not** a new technique. I do not claim the method is novel.

3. "Content guards are stronger on semantics" — is the whole stack weaker on semantics? (p14 underline)

No — the opposite. Inside Vyuha the surface L1 detector and the QLoRA guard are intentionally weak on semantics (L1 flags only **6.8%** of real PAIR jailbreaks). But the L2 content guard **Qwen3Guard carries semantics at 90.3% on PAIR and 79% on XSTest**, at 4.8% over-refusal. So the stack's semantic coverage equals Qwen3Guard's, which is strong.

The sentence means: rather than train our *own* semantic model to compete with mature content guards, we **compose the best existing one** and route to it through the selective cascade. The honest boundary is narrower than "weaker": Vyuha does not *add* semantic capability beyond the composed guard — its semantic ceiling is Qwen3Guard's. That is a composition boundary, not a coverage weakness.

4. Novelty — "highlight the novelty of this (L2)" and "are all operations borrowed or proposed?" (p7, p8, p2)

No component of Vyuha is a new algorithm. The borrowed, cited pieces include Dual-LLM (Willison), CaMeL as the target (DeepMind), Qwen3Guard, QLoRA, PSI drift monitoring, canary tokens, and the PAIR / AgentDojo / XSTest benchmarks.

On **L2 specifically** (your "highlight novelty of this" note): this is the layer with the *least* novelty. Both models are existing. L2's only defensible contributions are compositional — **selective invocation** (the expensive guard runs only on the cascade's uncertain band, which is what makes it affordable on free compute) and **measured complementarity** (we quantify that the content guard owns semantics while the tuned guard owns OOD jailbreak generalisation). I would not manufacture a novelty claim at L2; the numbers will not support one.

Where the closer-to-genuine contributions actually are (all measured, not asserted):

- **L0 adaptive multi-view de-spacing** — closed the character-spacing gap from ASR **0.83 → 0.00** and *generalised to a held-out wider-spacing variant it never trained on*. This generalisation is the strongest single novelty claim.
- **The augmentation-vs-normalisation ablation (C4)** — measured on 150 real in-the-wild seeds: normalisation alone leaves an adaptive attacker at **1.00 [95% CI 0.97-1.00]** ASR (a +0.98 premium over the static per-mutator number), and adversarial augmentation is what closes it to **0.03 [0.01-0.08] at 1% benign false positives**.
- **The selective-cascade division of labour, reproducible on a single free GPU** — the "which layer catches what" evidence.

So the honest framing is that **Vyuha is a systems, measurement, and reproducibility contribution, not an algorithmic one.**

5. New measured results since your review

Beyond the comments above, four evaluations were added since you reviewed the document — each with **95% confidence intervals**, now collected in **§7.4** of the updated Technical Document. They are the measured evidence behind the novelty framing:

- **NIST-AI-RMF guard benchmark** (reproducing arXiv:2605.28830): our 0.6B content guard reaches **0.799** recall on the complete-harmful-request axes — level with the benchmark's *best* 4B model (0.840) at $\sim 7\times$ smaller. This is standardised, external evidence that the composed guard is competitive, and it directly addresses the "is it performing well?" question.
- **Adaptive-attack robustness (C4)**: on 150 real seeds, normalisation alone is beaten by an adaptive attacker (ASR 1.00) but augmentation closes it to **0.03 at 1% FPR** — measured, not asserted.
- **Multi-turn / Crescendo detection**: session-level **0.71** versus **0.00** for per-message moderation.
- **Guard ensemble**: an OR union lifts recall $0.551 \rightarrow 0.715$, confirming the benchmark's ensembling recommendation (with an honest false-positive caveat).

The honest framing is unchanged — a systems / measurement / reproducibility contribution — but there is now more external, standardised evidence behind it.

Document edits made in response

1. **Page-13 chart** — added a second panel showing the tuned QLoRA guard on its own axis (OOD jailbreak ROC-AUC 0.72–0.92, a different metric from the detection rate), with a caption explaining the near-zero bars are by design.
2. **Executive summary (page 2)** — reworded to: "No single *component algorithm* is new... the contributions are the measured composition, the adaptive-robustness ablation, and reproducibility on free compute."
3. **New §7.4 (New measured results)** — a table of the four post-review evaluations above (NIST-RMF, adaptive C4, Crescendo, guard ensemble), each with 95% confidence intervals.